EX29 — Theory guide with a visual reference for counterclockwise rotation

This version of the theory guide adds a visual reference to interpret counterclockwise rotation in two dimensions. The goal is to clearly see what a positive angle means and how the rotation matrix acts on a vector.

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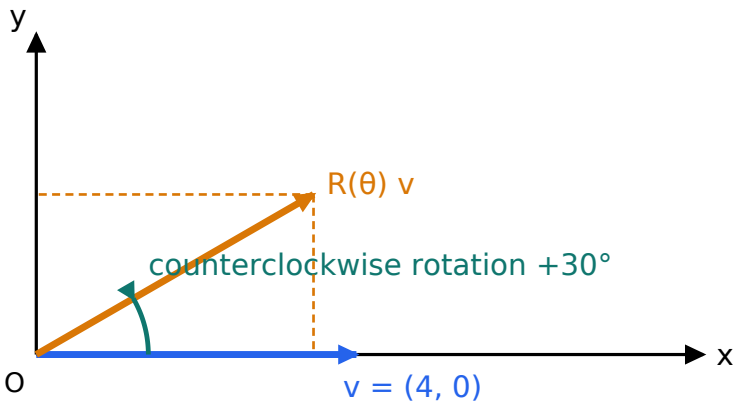
1. Rotation matrix in 2D

An active counterclockwise rotation of a vector in the plane is represented by:

$$R(\theta) = [[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$$

If θ is positive, the rotation is counterclockwise. If θ is negative, the rotation is clockwise.

2. Visual reference for counterclockwise rotation



Positive direction: opposite to the clock hands

Geometric example: the vector v = (4, 0) is rotated by +30° counterclockwise about the origin. The green arrow indicates the positive angle direction.

3. Numerical example

For the vector v = (4, 0) and θ = 30°, we obtain:

$$R(30°)\, v = (4\cos 30°, 4\sin 30°) = (3.4641, 2.0000)$$

The vector length does not change; only its direction changes. This is one of the fundamental properties of rotation matrices.

4. Two key ideas

• Rotating a vector means moving the vector inside the same coordinate system.
• Changing the reference frame means keeping the physical vector fixed and expressing its components in a different set of axes.

This distinction will be essential later when studying 3D rotations, Euler matrices, and changes between body, navigation, and inertial frames.